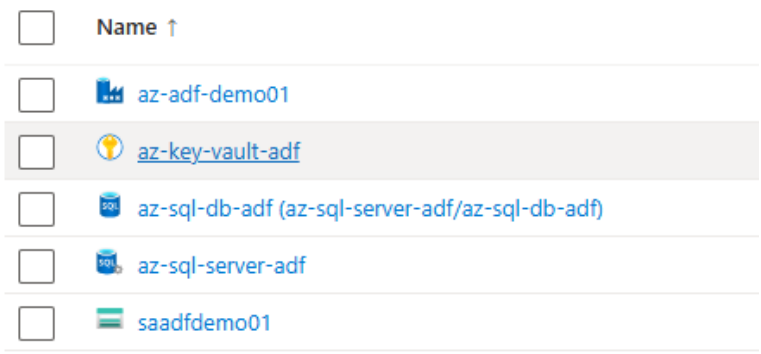
Name: Sumit Chauhan

Dept: Data Engineering

Module: ADF: Key Features, Use Cases, Key Terms

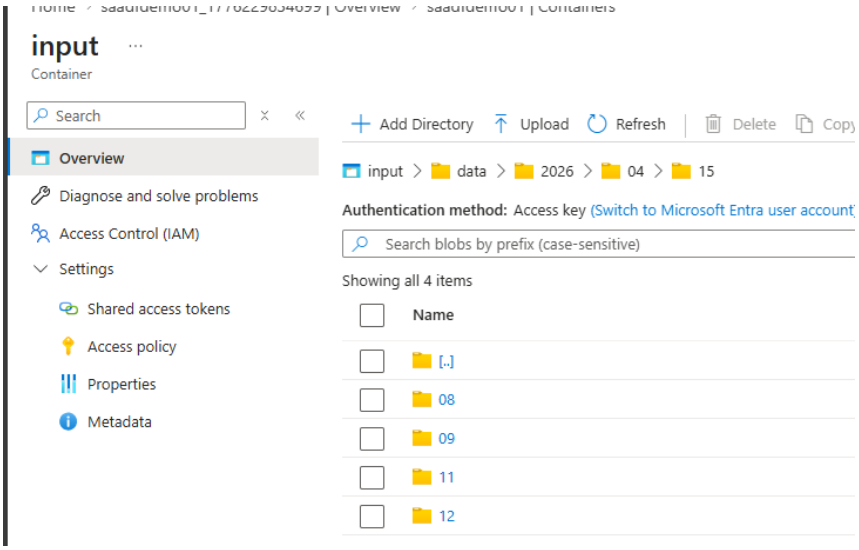
Task:

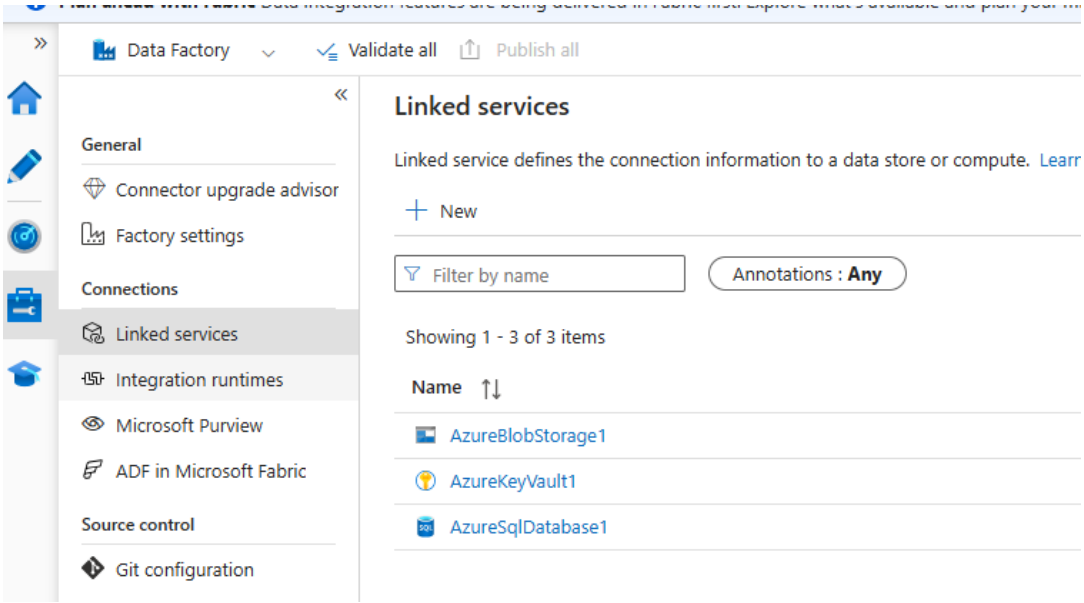


You storage account container (input) should look like this

Folder structure as dates and in each dates there is hour folder

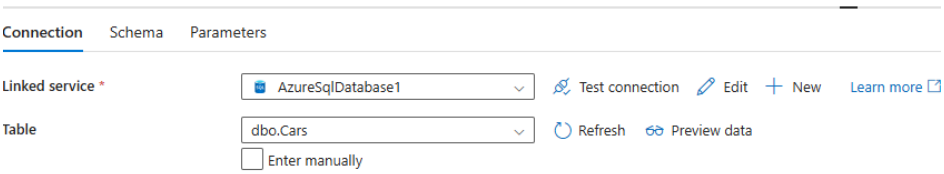
And in each hour folder there is csv file having updated content for that hour

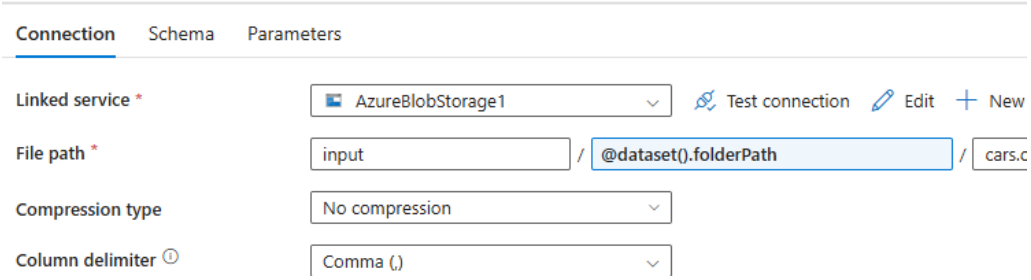




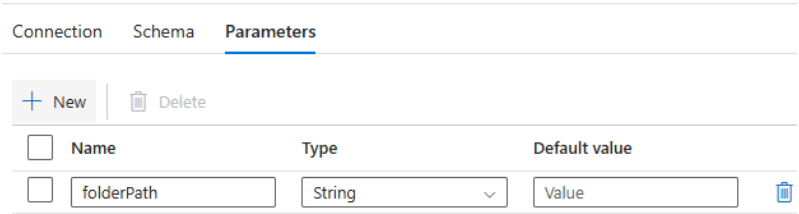
Now create datasets for sql database and blob storage

In blob storage keep folder Name as dynamic parameter passed at run time by pipeline





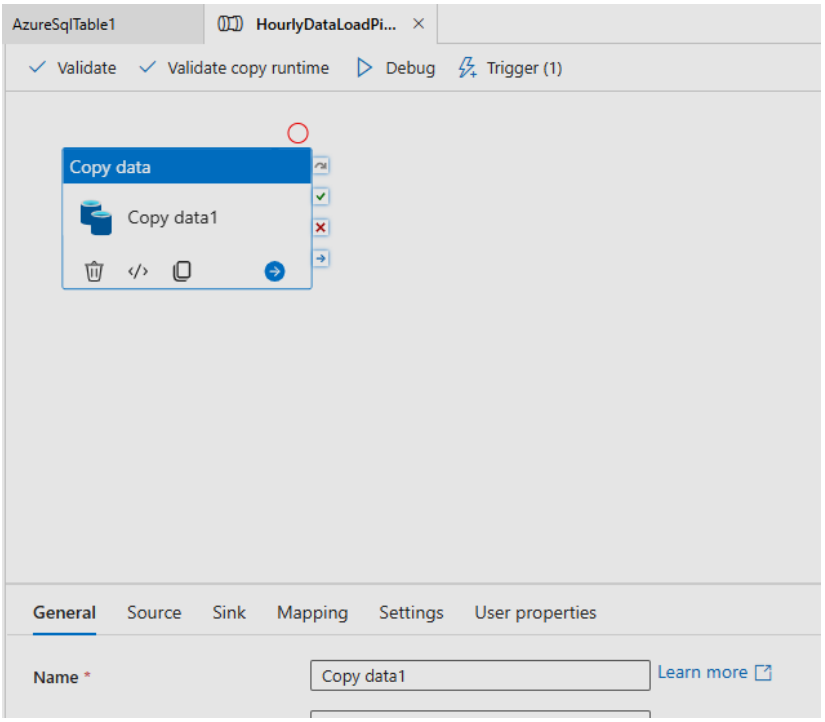
[note that here we have fixed the table and blob names, we can make them dynamic as well]



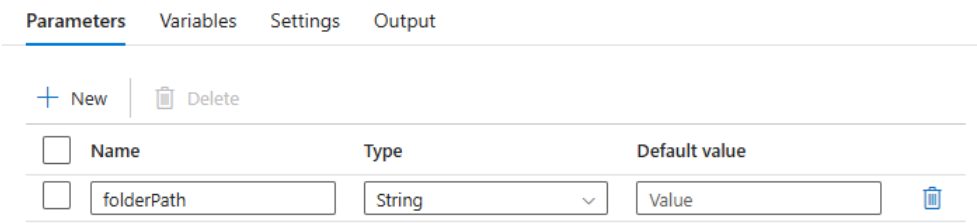
Now make HourlyPipeline and attach Source dataset as blob

And Sink dataset as sql database

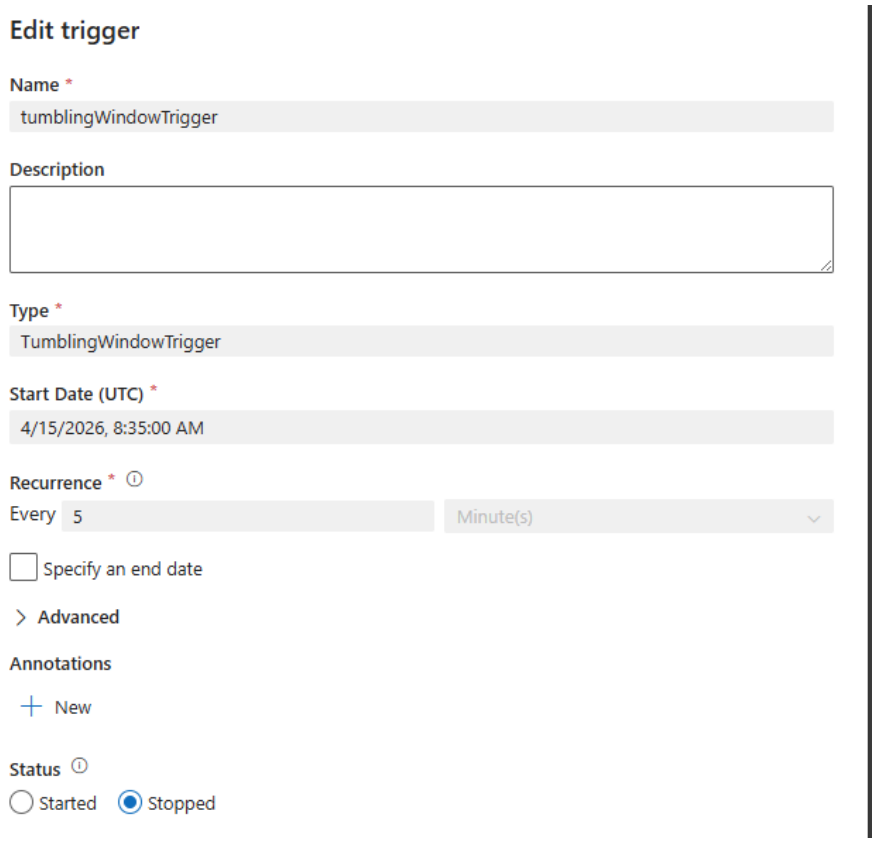
And add Copy Activity



Create a parameter for passing filePath to blob dataset

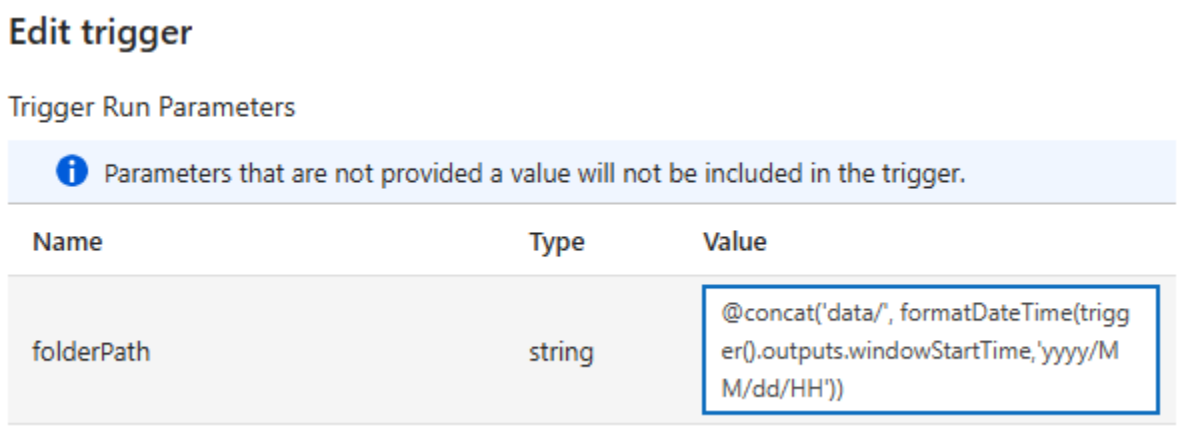


Create and attach tumblindWindowTrigger



Here we have taken recurrence every 5 minutes for learning purpose

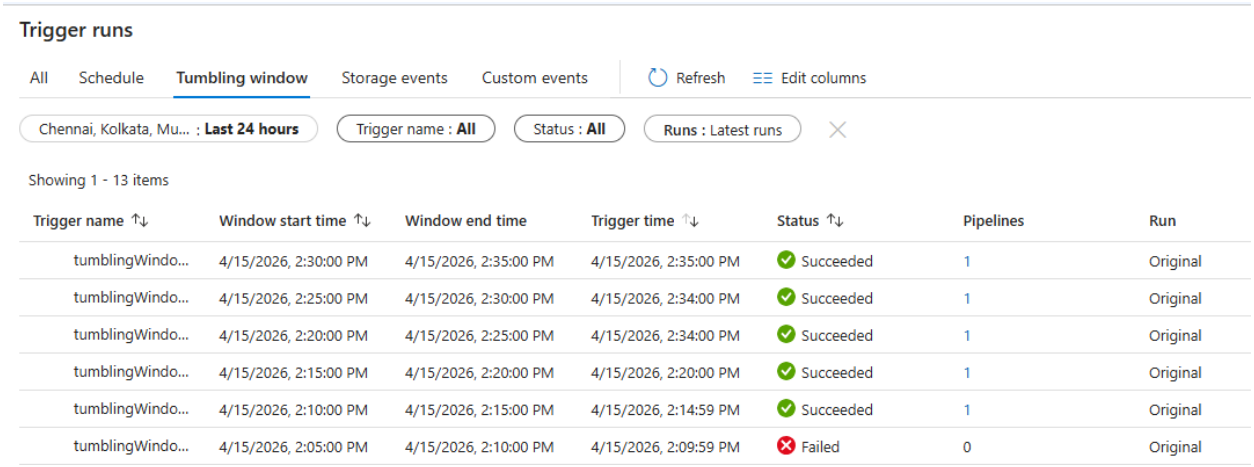
When you hit OK, you prompt to pass the parameters of pipeline



Carefully add the path like this

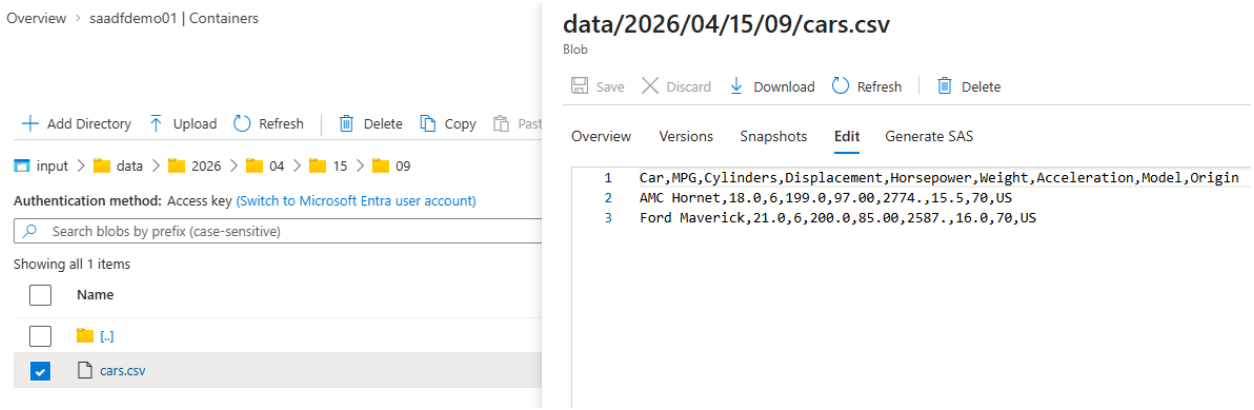
This will make a folderPath as per current date and current window

You can go to monitor tab to see the Trigger runs



Trigger name	Window start time	Window end time	Trigger time	Status	Pipelines	Run
tumblingWindo...	4/15/2026, 2:30:00 PM	4/15/2026, 2:35:00 PM	4/15/2026, 2:35:00 PM	Succeeded	1	Original
tumblingWindo...	4/15/2026, 2:25:00 PM	4/15/2026, 2:30:00 PM	4/15/2026, 2:34:00 PM	Succeeded	1	Original
tumblingWindo...	4/15/2026, 2:20:00 PM	4/15/2026, 2:25:00 PM	4/15/2026, 2:34:00 PM	Succeeded	1	Original
tumblingWindo...	4/15/2026, 2:15:00 PM	4/15/2026, 2:20:00 PM	4/15/2026, 2:20:00 PM	Succeeded	1	Original
tumblingWindo...	4/15/2026, 2:10:00 PM	4/15/2026, 2:15:00 PM	4/15/2026, 2:14:59 PM	Succeeded	1	Original
tumblingWindo...	4/15/2026, 2:05:00 PM	4/15/2026, 2:10:00 PM	4/15/2026, 2:09:59 PM	Failed	0	Original

Last added blob:



Final sql table:

Explorer

Search

az-sql-db-adf

dbo

Queries

SQL query 1

SQL query 1

Run | Save as view

```sql
CREATE TABLE Cars (
    Car varchar(80),
    MPG decimal(18,3),
    Cylinders decimal(18,3),
    Displacement decimal(18,3),
    Horsepower decimal(18,3),
    Weight decimal(18,3),
    Acceleration decimal(18,3),
    Model varchar(100),
    Origin varchar(100)
);

select * from Cars;
```

Messages | Results

Search

	Car	MPG	Cylinders	Displacement	Horsepower	Weight	Acceleration	Model	Origin
1	Datsun PL510	27.000	4.000	97.000	88.000	2130.000	14.500	70	Japan
2	Volkswagen 1131 Del...	26.000	4.000	97.000	46.000	1835.000	20.500	70	Europe
3	Chevrolet Chevelle M...	18.000	8.000	307.000	130.000	3504.000	12.000	70	US
4	Buick Skylark 320	15.000	8.000	350.000	165.000	3693.000	11.500	70	US
5	Plymouth Satellite	18.000	8.000	318.000	150.000	3436.000	11.000	70	US
6	AMC Hornet	18.000	6.000	199.000	97.000	2774.000	15.500	70	US
7	Ford Maverick	21.000	6.000	200.000	85.000	2587.000	16.000	70	US